

Polynomials: Division, Remainder and Factor Theorems



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To divide by $(x - a)$, put $x = a$ in: the answer is the remainder. If it is zero, $(x - a)$ is a factor — divide it out and factorise the quadratic that is left.

THE TOPIC ON ONE MAP

Polynomials: Division, Remainder and Factor Theorems

01 — Algebraic division

long division

quotient, remainder

missing terms $0x^2$

02 — Remainder theorem

$f(a)$

$f(\frac{b}{a})$ for $(ax - b)$

two equations

03 — Factor theorem

$f(a) = 0$

factorise fully

how many real roots?

ONE NUMBER DOES EVERYTHING

divide by $(x - a) \Rightarrow$ remainder $= f(a)$ divide by
 $(ax - b) \Rightarrow$ remainder $= f\left(\frac{b}{a}\right)$
 remainder 0 $\iff$ factor

01 Algebraic Division: Quotient and Remainder

Long division, one term at a time: divide the first terms, multiply back, subtract, bring down. Write every missing power as $0x^k$ first. The result is $f(x) = (\text{divisor}) \times Q(x) + R$.

WHEN YOU SEE $\rightarrow$ YOU DO

“find the quotient when $f(x)$ is divided by $(x - 2)$ ”

$\rightarrow$ long division (or compare coefficients)

“in the form $(x - 2)(ax^2 + bx + c) + d$ ”

$\rightarrow$ quotient gives a, b, c ; remainder is d

a missing term, e.g. $x^3 - 19x - 30$

$\rightarrow$ write $x^3 + 0x^2 - 19x - 30$

“hence, or otherwise”

$\rightarrow$ the remainder theorem gives R in one line

WORKED MODEL · JUNE 2026

$f(x) = 2x^3 - 7x^2 + 5x + 4$. Remainder and quotient when divided by $(x - 2)$.

1 $f(2) = 16 - 28 + 10 + 4 = 2$

remainder theorem

2 $2x^3 - 7x^2 + 5x + 4 = (x - 2)(2x^2 - 3x - 1) + 2$

long division

ANSWER $Q(x) = 2x^2 - 3x - 1, \quad R = 2$

IDEA 01

Practise: Algebraic Division: Quotient and Remainder

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Long division

P2 · TOPIC 2 3 marks

Divide $2x^3 + 3x^2 - 5x + 6$ by $(x + 2)$. State the quotient and the remainder.

Q2 DR ESLAM Write in the form

P2 · TOPIC 2 3 marks

Write $x^3 - 7x + 3$ in the form $(x - 2)(ax^2 + bx + c) + d$.

A missing x^2 term: write it as $0x^2$.

SELECTED PRACTICE

Q3

PRACTICE

Practice · Easy

P2 · TOPIC 2 4 marks

2 Divide $x^3 + 6x^2 + 11x + 6$ by $(x + 2)$.

(4 marks)**Q4**

PRACTICE

Practice · Medium

P2 · TOPIC 2 5 marks

6 Consider the function $f(x) = 4x^3 + 6x^2 - 7x + 2$.

(i) Find the quotient and remainder when $4x^3 + 6x^2 - 7x + 2$ is divided by $(x - 2)$.

(ii) Hence write $4x^3 + 6x^2 - 7x + 2$ in the form $(x - 2)(ax^2 + bx + c) + d$, where a , b , c and d are constants to be determined.

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 2 3 marks

1 Divide $4x^4 - 37x^2 + 9$ by $(2x - 1)$.

(3 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · January 2026 · Q3

P2 · TOPIC 2 7 marks

3. (i) Given that

- $f(x) = 4x^3 + 6x + k$, where k is a constant
- $(x + 2)$ is a factor of $f(x)$

find the remainder when $f(x)$ is divided by $(x - 5)$

(4)

(ii) Find the remainder, R , and the quotient, $Q(x)$, when

$$6x^3 - 15x^2 - 21x + 8$$

is divided by $(2x + 3)$

(3)

(Total for Question 3 is 7 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q1

P2 · TOPIC 2 4 marks**1.**

$$f(x) = 2x^3 - 7x^2 + 5x + 4$$

- (a) Use the remainder theorem to find the remainder when $f(x)$ is divided by $(x - 2)$ (2)
- (b) Hence, or otherwise, find the quotient when $f(x)$ is divided by $(x - 2)$ (2)

(Total for Question 1 is 4 marks)

02 The Remainder Theorem: Find the Unknowns

Each fact about a remainder or a factor gives one equation: put the root of the divisor into $f(x)$ and set it equal to the remainder. Two unknowns need two facts — then solve simultaneously.

WHEN YOU SEE → YOU DO

“when divided by $(x + 1)$ the remainder is 4”

$$\rightarrow f(-1) = 4$$

“ $(3x - 2)$ is a factor”

$$\rightarrow f\left(\frac{2}{3}\right) = 0$$

“show that $11A + 9B = 83$ ”

→ one fact → tidy into the given equation

“find the value of A and the value of B ”

→ two equations, solve simultaneously

“the remainder is k ” and “ $-10k$ ”

→ eliminate k between the two

Root of the divisor

Substitute into f

Set = remainder

Solve simultaneously

WORKED MODEL · JANUARY 2021

$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$. Remainder 4 on $(x + 1)$ and -23 on $(x - 2)$. Find a and b .

$$1 \quad f(-1) = 1 - a - 3 - b + 5 = 4 \Rightarrow a + b = -1$$

first fact

$$2 \quad f(2) = 16 + 8a - 12 + 2b + 5 = -23 \Rightarrow 4a + b = -16$$

second fact

ANSWER $a = -5, \quad b = 4$

× **TRAP** Using $f(3)$ for the factor $(2x - 3)$. The root of $2x - 3 = 0$ is $x = \frac{3}{2}$.

IDEA 02

Practise: The Remainder Theorem: Find the Unknowns

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Put the root in

P2 · TOPIC 2 2 marks

Find the remainder when $x^3 - 4x + 7$ is divided by (a) $(x - 3)$ (b) $(2x - 1)$.

Q9 DR ESLAM Two facts, two equations

P2 · TOPIC 2 5 marks

$f(x) = x^3 + ax^2 + bx - 6$. $(x - 1)$ is a factor, and the remainder on division by $(x - 2)$ is 18. Find a and b .

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 2**3 marks**

6 Given that $(x - 4)$ is a factor of $x^3 - kx^2 - 4x + 16$, find the value of k .

(3 marks)**Q11****PRACTICE**

Practice · Medium

P2 · TOPIC 2**9 marks**

5 (a) $f(x) = x^3 + 9x^2 + rx + s$. Given that $f(2) = 0$ and $f(-1) = -54$:
find the values of r and s .

(6 marks)

(b) Factorise $f(x)$ completely.

(3 marks)

Q12**PRACTICE**

Practice · Hard

P2 · TOPIC 2**6 marks**

- 6 Given that $(x + 4)$ is a factor of the function $f(x) = px^3 + (5p + 1)x^2 + 5qx - 2q - 2$ and that the remainder when $f(x)$ is divided by $(x + 1)$ is -12 , find the values of the constants p and q .

(6 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · January 2021 · Q1

P2 · TOPIC 2 6 marks

1. $f(x) = x^4 + ax^3 - 3x^2 + bx + 5$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b .

(4)

(Total 6 marks)

Q14 PAST PAPER Pearson IAL WMA12 · January 2022 · Q5

P2 · TOPIC 2 8 marks

5.
$$f(x) = 3x^3 + Ax^2 + Bx - 10$$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83 \quad (3)$$

Given also that $(3x - 2)$ is a factor of $f(x)$,

(b) find the value of A and the value of B . (3)

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x) \quad (2)$$

(Total 8 marks)

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q1

P2 · TOPIC 2 3 marks

1. $f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

(Total for Question 1 is 3 marks)

Q16**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q3

P2 · TOPIC 2 7 marks**3.**

$$f(x) = 2x^3 - x^2 + Ax + B$$

where A and B are integers.

Given that when $f(x)$ is divided by $(x + 3)$ the remainder is 55

(a) show that

$$3A - B = -118$$

(2)

Given also that $(2x - 5)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quotient when $f(x)$ is divided by $(x - 7)$

(2)

(Total for Question 3 is 7 marks)

Q17 PAST PAPER Pearson IAL WMA12 · June 2025 · Q4

P2 · TOPIC 2 9 marks

4: The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2 \quad (2)$$

Given further that when $f(x)$ is divided by $(2x - 1)$ the remainder is $\frac{7}{4}$

(b) find the value of a and the value of b . (4)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$ (3)

(Total for Question 4 is 9 marks)

03 The Factor Theorem: Factorise Fully and Count the Roots

Show $f(a) = 0$, so $(x - a)$ is a factor. Divide it out to get a quadratic, then factorise that (calculator first, then write the brackets). If the quadratic has $b^2 - 4ac < 0$, it gives no more real roots.

WHEN YOU SEE → YOU DO

“use the factor theorem to show $(2x + 3)$ is a factor”

→ $f(-\frac{3}{2}) = 0$, and say so

“hence fully factorise $f(x)$ ”

→ divide, then factorise the quadratic

“show that $f(x) = 0$ has only one real root”

→ quadratic factor: $b^2 - 4ac < 0$

“solve $6 \tan^3 \theta + 17 \tan^2 \theta + \dots = 0$ ”

→ same cubic in $\tan \theta$: roots, then angles

“using algebra”, “not calculator technology”

→ show the division and the factorised form

WORKED MODEL · JUNE 2023

$f(x) = 4x^3 - 8x^2 + 5x - 3$ with factor $(2x - 3)$. Show $f(x) = 0$ has only one real root.

1 $f(x) = (2x - 3)(2x^2 - x + 1)$

divide

2 $2x^2 - x + 1$: $b^2 - 4ac = 1 - 8 = -7 < 0$

no real roots

ANSWER only real root $x = \frac{3}{2}$

IDEA 03

Practise: The Factor Theorem: Factorise Fully and Count the Roots

My questions first, then selected practice, then real past papers.

Q18 **DR ESLAM** Factor, then factorise

P2 · TOPIC 2 4 marks

Show that $(x + 3)$ is a factor of $2x^3 + 5x^2 - 4x - 3$ and hence factorise it fully.

Q19 **DR ESLAM** Only one real root

P2 · TOPIC 2 3 marks

Show that $(x - 2)$ is a factor of $x^3 - 2x^2 + 2x - 4$, and that $x^3 - 2x^2 + 2x - 4 = 0$ has only one real root.

SELECTED PRACTICE

Q20

PRACTICE

Practice · Easy

P2 · TOPIC 2 2 marks

- 1 Use the factor theorem to verify that $(x - 2)$ is a factor of $x^3 - x^2 - 14x + 24$.

(2 marks)

Q21

PRACTICE

Practice · Medium

P2 · TOPIC 2 8 marks

- 4 (a) $f(x) = 4x^3 + 4x^2 - 23x - 30$

Use the factor theorem to show that $(x + 2)$ is a factor of $f(x)$.

(2 marks)

- (b) Factorise $f(x)$ completely.

(4 marks)

- (c) Write down all the real roots of the equation $f(x) = 0$.

(2 marks)

Q22**PRACTICE**

Practice · Very Hard

P2 · TOPIC 2 8 marks

5 (a) $f(x) = 2x^4 - 15x^3 - 10x^2 + 105x + 98$

Show that $f(-1) = 0$ and $f(-2) = 0$.

(1 mark)

(b) Hence, solve $f(x) = 0$.

(7 marks)

PAST PAPERS

Q23 PAST PAPER Pearson IAL WMA12 · October 2019 · Q4

P2 · TOPIC 2 8 marks

4. $f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

(a) State the remainder when $f(x)$ is divided by $(x - 3)$. (1)

Given $(3x - 2)$ is a factor of $f(x)$,

(b) show that $a = -17$ (2)

(c) Using algebra and showing each step of your working, fully factorise $f(x)$. (5)

(Total 8 marks)

Q24 PAST PAPER Pearson IAL WMA12 · January 2020 · Q3

P2 · TOPIC 2 8 marks

3. $f(x) = 6x^3 + 17x^2 + 4x - 12$

(a) Use the factor theorem to show that $(2x + 3)$ is a factor of $f(x)$. (2)

(b) Hence, using algebra, write $f(x)$ as a product of three linear factors. (4)

(c) Solve, for $\frac{\pi}{2} < \theta < \pi$, the equation

$$6 \tan^3 \theta + 17 \tan^2 \theta + 4 \tan \theta - 12 = 0$$

giving your answers to 3 significant figures. (2)

(Total 8 marks)

Q25**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q4

P2 · TOPIC 2 8 marks

4. $f(x) = (x^2 - 2)(2x - 3) - 21$

(a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$ (1)

(b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$ (2)

(c) Hence,

(i) factorise $f(x)$

(ii) show that the equation $f(x) = 0$ has only one real root. (5)

(Total 8 marks)

Q26**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q5

P2 · TOPIC 2 9 marks

5.
$$f(x) = x^3 + (p + 3)x^2 - x + q$$

where p and q are constants and $p > 0$

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that

$$9p + q = -51 \quad (2)$$

Given also that when $f(x)$ is divided by $(x + p)$ the remainder is 9

(b) show that

$$3p^2 + p + q - 9 = 0 \quad (2)$$

(c) Hence find the value of p and the value of q . (3)

(d) Hence find a quadratic expression $g(x)$ such that

$$f(x) = (x - 3)g(x) \quad (2)$$

(Total for Question 5 is 9 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q2

P2 · TOPIC 2 6 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where a is a constant.

Given that $(2x - 3)$ is a factor of $f(x)$,

(a) use the factor theorem to show that $a = -3$ (2)

(b) Hence show that the equation $f(x) = 0$ has only one real root. (4)

(Total for Question 2 is 6 marks)

Q28**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q4

P2 · TOPIC 2**9 marks**

4. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + ax^2 - 29x + b$$

where a and b are constants.

Given that $(2x + 1)$ is a factor of $f(x)$,

(a) show that

$$a + 4b = -56 \quad (2)$$

Given also that when $f(x)$ is divided by $(x - 2)$ the remainder is -25

(b) find a second simplified equation linking a and b . (2)

(c) Hence, using algebra and showing your working,

(i) find the value of a and the value of b ,

(ii) fully factorise $f(x)$. (5)

(Total for Question 4 is 9 marks)

Q29**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q4

P2 · TOPIC 2 8 marks

4.

$$f(x) = (x - 2)(2x^2 + 5x + k) + 21$$

where k is a constant.

- (a) State the remainder when $f(x)$ is divided by $(x - 2)$

(1)

Given that $(2x - 1)$ is a factor of $f(x)$

- (b) show that $k = 11$

(2)

- (c) Hence

(i) fully factorise $f(x)$,

(ii) find the number of real solutions of the equation

$$f(x) = 0$$

giving a reason for your answer.

(5)

(Total for Question 4 is 8 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q5

P2 · TOPIC 2 9 marks**5.**

$$f(x) = 3x^3 + ax^2 - 10x + b$$

where a and b are constants.

Given that $(3x - 4)$ is a factor of $f(x)$,

(a) show that $16a + 9b = 56$

(2)

Given further that when $f(x)$ is divided by $(x - 2)$ the remainder is b ,

(b) find the value of a and the value of b .

(4)

(c) Hence, using algebra, fully factorise $f(x)$.

(3)

(Total for Question 5 is 9 marks)

Q31**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q4

P2 · TOPIC 2 12 marks

4. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 + 13x^2 - 10x + 8$$

- (a) When $f(x)$ is divided by $(x - 2)$ the remainder is R and the quotient is $Q(x)$.

(i) Find $Q(x)$.

(ii) Find R .

(4)

- (b) (i) Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$.

(ii) Hence prove, using algebra, that the equation $f(x) = 0$ has only one real solution.

(5)

- (c) Find the range of values of x for which $f(x)$ is decreasing.

(3)

(Total for Question 4 is 12 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Division** $0x^k$ for gaps; $f = (\text{divisor})Q + R$.
- 02 Remainder theorem** Root of the divisor into f ; one fact = one equation.
- 03 Factor theorem** $f(a) = 0$; divide; factorise; $b^2 - 4ac < 0 \rightarrow$ no more roots.

BOTTOM LINE Root in $\rightarrow$ remainder $\cdot$ zero $\rightarrow$ factor $\rightarrow$ divide $\rightarrow$ factorise the quadratic.

Answers

Polynomials: Division, Remainder and Factor Theorems · for the teacher.
Every answer checked with a computer algebra system.

IDEA 01 · ALGEBRAIC DIVISION: QUOTIENT AND REMAINDER

- Q1** Dr Eslam · Long division quotient $2x^2 - x - 3$, remainder 12
- Q2** Dr Eslam · Write in the form $(x - 2)(x^2 + 2x - 3) - 3$: $a = 1$, $b = 2$, $c = -3$, $d = -3$
- Q3** Practice · Easy $x^2 + 4x + 3$ (remainder 0)
- Q4** Practice · Medium (i) quotient $4x^2 + 14x + 21$, remainder 44 (ii) $a = 4$, $b = 14$, $c = 21$, $d = 44$
- Q5** Practice · Very Hard $2x^3 + x^2 - 18x - 9$, remainder 0
- Q6** Pearson IAL WMA12 · January 2026 · Q3 (i) 574. (ii) $Q(x) = 3x^2 - 12x + \frac{15}{2}$, $R = -\frac{29}{2}$
- Q7** Pearson IAL WMA12 · June 2026 · Q1 (a) $f(2) = 2$ (b) $2x^2 - 3x - 1$

IDEA 02 · THE REMAINDER THEOREM: FIND THE UNKNOWN

- Q8** Dr Eslam · Put the root in (a) $f(3) = 22$ (b) $f(\frac{1}{2}) = \frac{41}{8}$
- Q9** Dr Eslam · Two facts, two equations $a + b = 5, 2a + b = 8: a = 3, b = 2$
- Q10** Practice · Easy $f(4) = 64 - 16k = 0 \Rightarrow k = 4$
- Q11** Practice · Medium (a) $r = 6, s = -56$ (b) $(x - 2)(x + 4)(x + 7)$
- Q12** Practice · Hard $f(-4) = 0: 8p - 11q = -7; f(-1) = -12: 4p - 7q = -11; p = 6, q = 5$
- Q13** Pearson IAL WMA12 · January 2021 · Q1 (a) Proven: $a + b = -1$. (b) $a = -5, b = 4$
- Q14** Pearson IAL WMA12 · January 2022 · Q5 (a) Proven: $11A + 9B = 83$. (b) $A = -8, B = 19$. (c) $g(x) = x^2 - 2x + 5$
- Q15** Pearson IAL WMA12 · January 2024 · Q1 $a = \frac{5}{8}$
- Q16** Pearson IAL WMA12 · October 2024 · Q3 (a) Proven. (b) $A = -26, B = 40$. (c) $2x^2 + 13x + 65$
- Q17** Pearson IAL WMA12 · June 2025 · Q4 (a) Proven. (b) $a = 2, b = 8$. (c) Quotient $2x^2 + 12x + 29$, remainder 55

IDEA 03 · THE FACTOR THEOREM: FACTORISE FULLY AND COUNT THE ROOTS

- Q18** Dr Eslam · Factor, then factorise
 $f(-3) = 0; (x + 3)(2x + 1)(x - 1)$
- Q19** Dr Eslam · Only one real root
 $(x - 2)(x^2 + 2); x^2 + 2$ has discriminant $-8 < 0$, so $x = 2$ only
 $f(2) = 8 - 4 - 28 + 24 = 0$, so $(x - 2)$ is a factor
- Q20** Practice · Easy
 (a) $f(-2) = 0$ (b) $(x + 2)(2x - 5)(2x + 3)$ (c) $x = -2, \frac{5}{2}, -\frac{3}{2}$
- Q21** Practice · Medium
 (a) both give 0 (b) $(x + 1)(x + 2)(x - 7)(2x - 7): x = -2, -1, \frac{7}{2}, 7$
- Q22** Practice · Very Hard
 (a) -35 . (b) $a = -17$. (c) $f(x) = (3x - 2)(x + 2)(x - 4)$
- Q23** Pearson IAL WMA12 · October 2019 · Q4
 (a) Proven. (b) $f(x) = (2x + 3)(3x - 2)(x + 2)$. (c) $\theta = 2.03, 2.16$ radians
- Q24** Pearson IAL WMA12 · January 2020 · Q3
 (a) -21 . (b) Proven. (c)(i) $f(x) = (x - 3)(2x^2 + 3x + 5)$. (c)(ii) Only real root: $x = 3$
- Q25** Pearson IAL WMA12 · October 2021 · Q4
 (a) Proven. (b) Proven. (c) $p = 6, q = -105$. (d) $g(x) = x^2 + 12x + 35$
- Q26** Pearson IAL WMA12 · January 2023 · Q5
 (a) $a = -3$. (b) $f(x) = 0$ has only one real root, $x = \frac{3}{2}$
- Q27** Pearson IAL WMA12 · June 2023 · Q2
 (a) Proven. (b) $4a + b = 1$. (c)(i) $a = 4, b = -15$. (c)(ii) $f(x) = (2x + 1)(x - 2)(2x + 15)$
- Q28** Pearson IAL WMA12 · October 2023 · Q4
 (a) 21. (b) $k = 11$. (c)(i) $(2x - 1)(x^2 + x + 1)$. (c)(ii) One real solution
- Q29** Pearson IAL WMA12 · June 2024 · Q4
 (a) Proven. (b) $a = -1, b = 8$. (c) $f(x) = (x - 1)(x + 2)(3x - 4)$
- Q30** Pearson IAL WMA12 · January 2025 · Q5
 (a)(i) $Q(x) = 4x^2 + 21x + 32$. (a)(ii) $R = 72$. (b) Proven; only real solution is $x = -4$. (c) $-\frac{5}{2} < x < \frac{1}{3}$
- Q31** Pearson IAL WMA12 · October 2025 · Q4